

Fabiano Cacioli

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PROFILE

MSc student in Computer Engineering and Artificial Intelligence (Sapienza, taught in English), with hands-on experience in **applied cryptography and security**, **machine learning**, and full-stack development. Seeking a first internship or junior role in cybersecurity (open to broader software and data roles) to apply these skills and grow within a team.

EDUCATION

Sapienza University of Rome — Rome, IT

Oct 2025 – Present (exp. Oct 2027)

MSc in Computer Engineering and Artificial Intelligence

Relevant coursework: Cybersecurity, Machine Learning, Artificial Intelligence, Advanced Programming, Interactive Graphics, Cloud Computing.

Università degli Studi Roma Tre — Rome, IT

Oct 2022 – Jul 2025

BSc in Economics and Big Data — Final grade: 101/110

TECHNICAL PROJECTS

Applied Cryptography & Security — Cybersecurity course

- Implemented and benchmarked symmetric ciphers (AES-128, ChaCha20, Camellia, SM4) and authenticated-encryption schemes (AES-GCM, ChaCha20-Poly1305, Encrypt-then-MAC) in C++ and Python using OpenSSL and the cryptography library.
- Built and statistically compared two CSPRNG/DRBG designs (ChaCha20 stream vs SHA-256 Hash-DRBG, per NIST SP 800-90A) and implemented a secure commit–reveal protocol over TCP sockets across two Linux VMs.
- Performed classical cryptanalysis (frequency analysis) to break a monoalphabetic substitution cipher.
- **Tools:** C++, Python, OpenSSL, Linux, sockets, VirtualBox.

Machine Learning: Dimensionality Reduction & Deep Learning (team project)

- Compared PCA vs autoencoders across three tasks — K-Means clustering (HAR), FNN regression (California Housing) and CNN classification (Fashion-MNIST).
- CNN reached ~89.5% accuracy on Fashion-MNIST; a 75% feature reduction cost only ~4% accuracy, quantifying the compute/accuracy trade-off.
- **Tools:** Python, scikit-learn, TensorFlow/Keras, pandas, NumPy.

Financial Sentiment Analysis (NLP) (team project)

- Built a 3-class sentiment classifier on ~5,800 financial sentences using TF-IDF (+SVD), comparing Naive Bayes, Softmax, Linear SVM, Decision Tree and Random Forest with 5-fold cross-validation (macro-F1), plus confusion-matrix, ROC/AUC and feature-importance analysis.

Seismic Event Detection Web App — Hackathon (Advanced Programming)

- Built a working full-stack application (frontend + backend) to detect and report seismic events, delivered as a team within a hackathon.

Real-Time 3D Graphics — Interactive Graphics

- Implemented a WebGL mesh renderer with Blinn–Phong shading, model/view transformations and texture mapping in JavaScript and GLSL shaders.

BSc Thesis — Privacy, Data Protection & GDPR

- Researched GDPR and data-subject rights applied to public-sector entities — a foundation directly relevant to security governance, privacy and compliance roles.

TECHNICAL SKILLS

Programming: Python, C/C++, Java, JavaScript (incl. GLSL), SQL, R

Cybersecurity & Cryptography: AES, ChaCha20, AEAD (GCM, Poly1305), HMAC, HKDF/scrypt, CSPRNG/DRBG, commitment schemes, OpenSSL, age, GDPR/data protection

Machine Learning & Data: scikit-learn, pandas, NumPy, Matplotlib; neural networks (CNN, FNN, autoencoders); PCA/clustering; TF-IDF / NLP

Tools & Systems: Linux, Git, Docker, VirtualBox, TCP/socket programming, Google Colab, LaTeX

Languages: Italian (native); English — B2 (entire MSc taught in English)